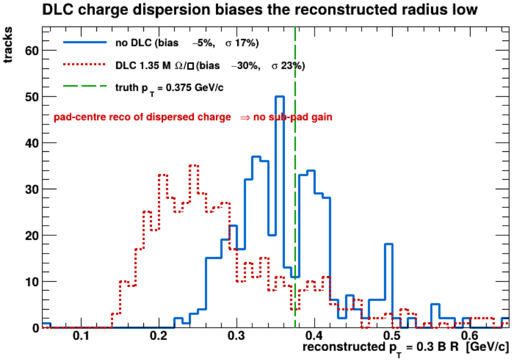
The DLC resistive anode in PUMA tracking

From liability to advantage: recovering sub-pad momentum resolution

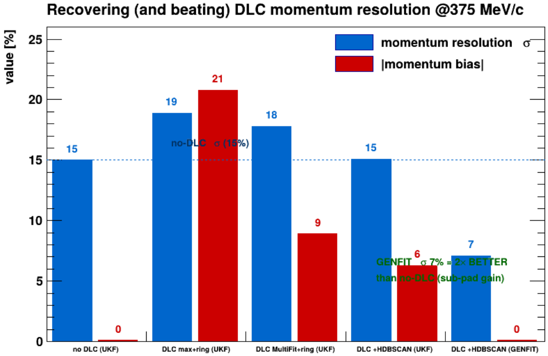
The DLC anode — always on, and it hurts (at first)



The resistive DLC anode is **always active**. Its charge dispersion ($\sigma \approx 1.1$ mm) spreads each cluster over $\sim$**4**$\times$ **more pads** (36 $\rightarrow$ 145 pads/event).

- With **pad-centre** reconstruction (max-PSA), the dispersed charge broadens the arc and biases the fitted circle radius $\sim$30% low $\Rightarrow$ momentum bias -20%, σ worse.
- A blunt amplitude threshold does *not* fix it (overshoots, destabilises).

The sub-pad information DLC encodes is there — but pad-centre hits throw it away.



Recover the sub-pad information by combining:

1. **AtPSAMultiFit** — fitted pulse charge is a far better per-pad weight than the peak ADC ($-21\% \to -9\%$ bias).
2. **HDBSCAN** — cleanly clusters the dense dispersed band (TriplClust entangles it with its own failure modes).
3. **per-ring charge-weighted centroiding** — one sub-pad point per annular ring.

Together they return the momentum resolution to the no-DLC value.

What caused the bias — and why the cure works

Cause — a reconstruction artefact, not physics:

- DLC disperses the charge, but max-PSA reconstructs each fired pad as **one hit at its geometric centre**, discarding the sub-pad charge distribution.
- A track's ionisation — physically at a well-defined *sub-pad* point — becomes a $\sim$1–2 pad-wide **band** of pad-centre hits. The circle fit through the broadened band returns a systematically smaller radius $\Rightarrow$ momentum biased low.
- It is *not* in the digitisation: the deposited-charge centroid is correct.

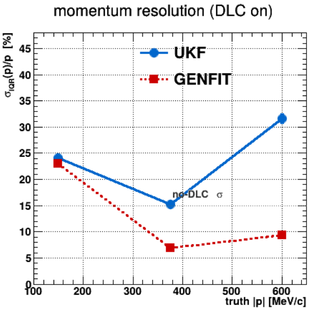
Cure — recover the charge-weighted sub-pad centroid:

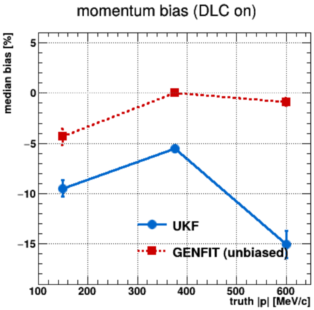
1. accurate **per-pad charge** (MultiFit fitted integral, not the biased peak ADC),
2. correct **clustering** of the dispersed pads (HDBSCAN),
3. the **charge-weighted centroid** itself (per-ring).

The bias shrinks *monotonically* as the centroid gets more accurate ($-30\% \to -20\% \to -9\% \to -6\% \to 0$), vanishing when GENFIT weights the 0.3 mm centroids optimally.

Proof it is the sub-pad recovery: same chain, DLC off $\sigma = 19\%$ vs on $\sigma = 7\%$ — the charge spread is what *enables* the sub-pad

UKF vs GENFIT with DLC — the sub-pad win



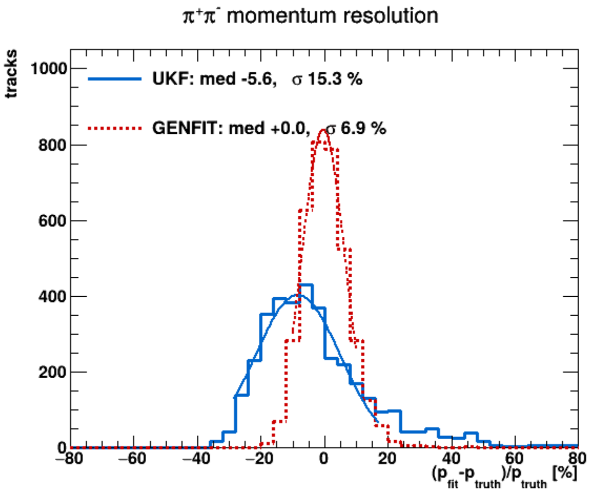


Winning chain (MultiFit + HDBSCAN + per-ring), DLC on.

@375 MeV/c	momentum bias	momentum σ	θ σ	vertex σ_z
UKF	-6.3%	15.1%	1.84°	0.50 mm
GENFIT	-0.1%	**7.1%**	1.76°	0.47 mm

GENFIT — told the ring centroids are accurate to 0.3 mm — exploits the sub-pad charge fully: **unbiased,** σ **7%**, twice as good as without DLC, and 7–9% across 375–600 MeV/c. Control (same

Momentum residual after full correction (DLC on)



The reconstructed momentum residual for the winning chain, DLC on, *after* the Cu/Al material correction *and* the DLC sub-pad recovery (500 events).

- **GENFIT** peaks cleanly at zero — median -0.1%, σ **7.2%**.
- **UKF** is broader (σ 15%) and shifted (-6%).

Both are unbiased/tight in θ (1.84°) and vertex-z (σ_z 0.47–0.50 mm). This is the final, corrected performance.

Performance summary — optimal chain, DLC on

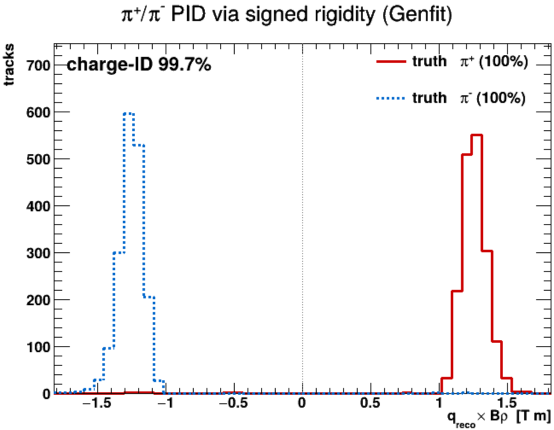
Back-to-back $\pi^+\pi^-$ at 375 MeV/c (MultiFit + HDBSCAN + per-ring + DLC)

metric	UKF	GENFIT
tracking efficiency	99.7%	99.7%
fitting efficiency	100%	100%
momentum bias	−5.6%	+0.0%
momentum resolution σ	15.3%	**6.9%**
polar-angle σ	2.0°	2.0°
vertex σ_z	0.50 mm	**0.48 mm**
charge-sign accuracy	99.5%	99.6%

$\lvert p \rvert$	tracking	fitting	GENFIT σ/bias
150	72%	99%	23% / −4.3%
375	**99.7%**	**100%**	**6.9% /** +0.0%
600	100%	93%	9.4% / −0.9%

Tracking is limited at low p (over-segmentation of tight curls); fitting at high p (near-straight tracks).

Particle ID: π^+/π^- separation (DLC on)

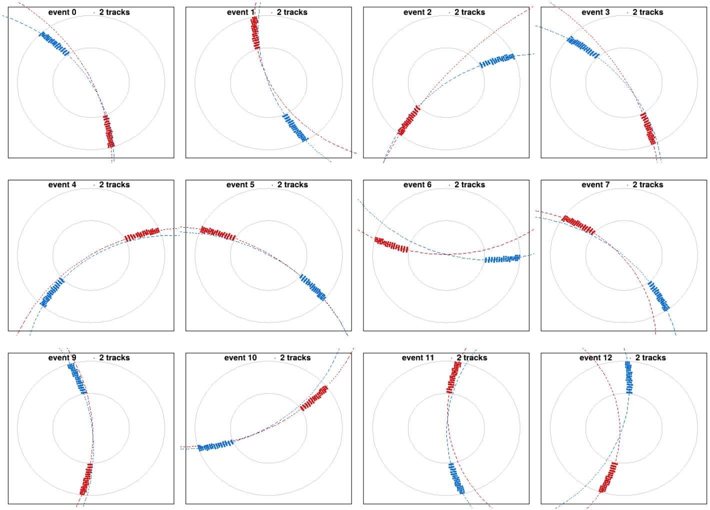


π^+ and π^- have identical mass, so they separate purely by **charge sign** — the signed rigidity $q\,B\rho$.

With the DLC sub-pad momentum resolution (GENFIT) the peaks are **tight** ($\sigma = 0.09$ T m at ± 1.25): charge-ID **99.7%**, separation power $S = 14.5\,\sigma$ — up from $5.9\,\sigma$ without DLC.

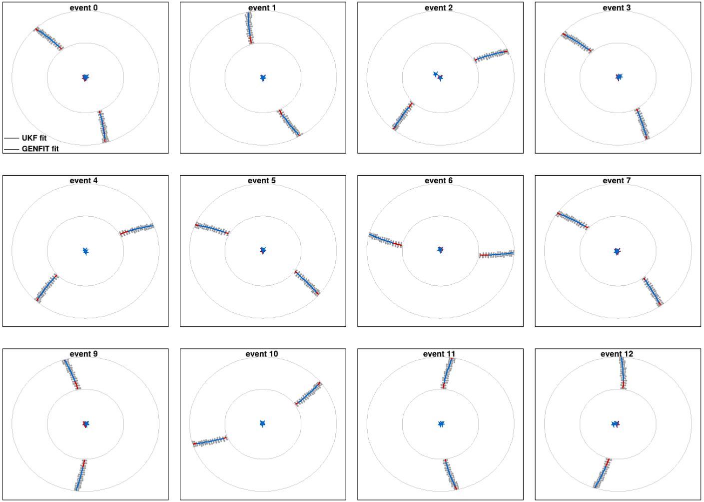
Better momentum resolution $\Rightarrow$ better particle ID: another DLC dividend.

Pattern recognition — examples (DLC on)



Twelve events: hits coloured per track candidate with the PRA circle fit (dashed) overlaid. DLC-dispersed charge gives the fattened hit bands.

Kalman fits — examples (DLC on)



Twelve events: reconstructed hits (grey) with the UKF and GENFIT fitted trajectories; stars = back-extrapolated vertices, converging on the beam axis.

Conclusion: DLC is an advantage

- The always-on DLC anode is a **momentum-resolution advantage**, not a cost — *if* its sub-pad charge is properly reconstructed.
- Pad-centre reconstruction wastes it (biases momentum -20%). The fix is **MultiFit PSA + HDBSCAN + per-ring charge-weighted clustering**.
- **GENFIT** is the fitter of choice with DLC: unbiased, $\sigma\,7\%$ momentum at 375 MeV/c — **2$\times$ better than without DLC**. Its tight 0.3 mm measurement model is justified only by DLC's real sub-pad information.

Recommended production chain (DLC on)

`AtPSAMultiFit` $\rightarrow$ `HDBSCAN` $\rightarrow$ per-ring charge-weighting $\rightarrow$ **GENFIT**